Defect workflow

Read each column from top to bottom, A to G: F and G share the last one. Diamonds are decisions, and every branch out of one is labelled; "given" means set in the project file. Dashed steps are optional and off the main path. One command runs every step in order: `python3 dwf.py <name>.dwf`. Each phase writes summary.txt (for you to read) and results.json (for the later phases) in its own folder.

start / end · done by dwf.py itself · CASTEP jobs (run through supercell_convergence.py) · MACE (B2, C1: a foundation model; D3–D5, G2: one trained per defect) · ShakeNBreak · needs your attention · decision · input or data · optional, off the main path · (B) continues in the next column

A Host settings and reference data

A_host/ · once per host

project file <name>.dwf, host .cell and .param, jobscript

Converge k-points and cut-off? (converge)
- no → Use your .param and k-points as they are
- yes

A1 Converge k-points and cut-off
host energy per atom on each k-point grid (at a low cut-off), then for each cut-off in converge_cutoffs

Has the energy settled, within converge_tol?
- no → Not settled: add finer grids or higher cut-offs, then rerun
- yes

A2.1 Relax the host
lattice and positions, at A1's settings; every later cell is built from it

A2.2 Band edges
a single point on a dense Γ-centred grid: VBM, CBM and band gap

Dielectric constant given? (dielectric)
- yes → Use that value
- no

A2.3 Dielectric constant
DFPT at Γ (phonon+efield): ε_0 and ε_∞; ε_0 screens every charge correction

Chemical potentials given?
- yes → Use them (%block chemical_potentials)
- no

Competing phases given? (phases, or phases_from)
- no → No limits: formation energies without μ terms
- yes

A3.1 Relax the competing phases
yours, or the Materials Project's (doped), at the same settings as the host

Is the host stable against them?
- no → Stop: check those phases and their settings
- yes

A3.2 Chemical-potential limits
μ at each corner of the host's stability region (pymatgen); each dopant's μ is the highest its own phases allow

(B)

B Candidates, the screening and electronic cells

B_candidates/ · once per host

(A)

Defects listed in %block defects?
- yes → **B1 Your defects** one candidate per line; a pair with all: every arrangement of it
- no

B1 List the candidates
vacancies and dopant substitutions on each inequivalent site (antisites and interstitials too, if switched on), then dopant–native pairs in every distinct arrangement up to pair_max_distance

B2 Supercell sizes
for each length in scan_lengths, the most compact supercell whose periodic images are at least that far apart (found by doped, or n × your transform)

Test each candidate's size with MACE? (mace and scan)
- no → No test: B3 goes by the Γ-only floor alone
- yes

B2 MACE size test
each candidate relaxed in every size it fits; it needs the smallest size whose relaxation energy matches every bigger size's within scan_tol, with its edge atoms moving less than scan_edge_tol

Screening cell given? (supercell)
- yes → Use it (B2's test is then only reported)
- no

B3 Choose the screening cell
the smallest size that is big enough for every candidate (the worst case decides) and at least the Γ-only floor (note below)

Do the host's band edges (A2.2) fold onto Γ of this cell?
- no → Warning: Γ-point screening sees a wider gap (C2 measures it); B4's electronic cell folds the edges
- yes

B4 The electronic cell (C4 on)
the smallest size, from the screening cell up, that folds every band-edge valley onto Γ (or electronic_cell)

(C)

The Γ-only floor (B3)

C screens at the Γ point only, which gives a defect's levels as an isolated defect has them only when its periodic images barely overlap. A level E_b inside the gap decays as exp(−r/ξ), where (two-band model) $1/\xi^2 = (2m^*/\hbar^2)\,E_b\,(1 - E_b/E_g)$. The screening cell keeps its images at least 4ξ apart, with
- E_b = deep_level_margin: the shallowest level that counts as deep
- m*: the lighter of the hole and electron masses at A2.2's band edges
- E_g: A2.2's band gap

Set screen_min_image to give the floor yourself (0: no floor).

Cell sizes are image distances

The shortest distance between a defect and its periodic images (Å), from scan_lengths: B2 tests every candidate in every size, B3's screening cell fits the worst case and the Γ-only floor, and B4's electronic cell (C4 on) also folds the band edges onto Γ.

C Screening

C_screen/ · every candidate, then the survivors in the electronic cell

(B)

C1 Build and pre-relax
every candidate in the screening cell, pre-relaxed with MACE (reusing B2's)

Filter by MACE energy? (configuration_margin)
- yes → Drop arrangements more than that above the lowest MACE energy of their composition
- no

C2 Screen in CASTEP
a cheap Γ-point relaxation of every candidate in each of screen_charges, from a triplet (a doublet for an odd electron count), plus the pristine cell

C3: passes the hull (charge-corrected), duplicate and property filters?
- no → Dropped (the reason is in summary.txt)
- yes

Into the electronic cell (B4)
everything after C3 works in it: each survivor's screened structures carried over (distort_cell) and relaxed in C5

C4 Distort (ShakeNBreak)
in the electronic cell, each charge state C3 carries on: neighbours moved in and out by each of distortions, others rattled

C5 Relax the distortions
and each start, at Γ with C2's settings, against the electronic cell's band edges; every step kept: D1's training data

(D)

What C keeps for D1–D5

Every geometry step of C5, with its energy and forces, in each .castep: D1 turns them into each defect's training data (the pbe head, at the screening settings), and D2 recomputes a tenth of them with the hybrid, in the electronic cell like everything after C3. With model : false, D6 ranks C5's relaxations instead.

The model in D1–D5: Wang et al.

Wang, Mosquera-Lois and Walsh, PRX Energy 2026 (arXiv:2603.05238): Sb_2Se_3, one 60-atom cell, VASP (PAW, 350 eV, DFT-D3).
- $V_{Se}(1)$ in q = −2 … +2: one charge-embedded MACE, trained on HSE06 Γ-point relaxations of ShakeNBreak distortions (2,287 structures)
- neutral $V_{Se}(1)$: a multi-fidelity MACE from 452 PBE steps and 34 HSE06 single points
- MACE 0.3.15 from scratch: 2 layers, 128 channels, 5.5 Å cut-off, ZBL
- test errors 0.5 meV/atom and 20 meV/Å; transition levels within 0.015 eV of HSE06; the search about 1,000 × cheaper than HSE06
- one model per defect, site and cell size; negative charge states were the hardest; the charge embedding and the two heads (D3) were never tested together: they trained them as two models

Data and trained models: Zenodo, record 22756145 (CC BY 4.0).

Foundation models on defects (their test)

On $V_{Sb}(1)$ at the PBE level, MACE-MH-1, MACE-MPA-0, GRACE-2L-OMAT and MatterSim put the minima 0.2–0.4 Å from DFT's (0.1 Å counts as a match), on flatter energy surfaces, and DFT relaxations from their minima missed the DFT ground states. The charge-aware MACE-POLAR-1-L and QET-PES-MatQ failed too. So B2 and C1 keep a foundation MACE for cell sizes and pre-relaxation only, and each defect gets a model trained on its own DFT data (D3).

What the filters in C and D keep
- C1 configuration_margin: arrangements within that of the lowest MACE energy of their composition (MACE ignores the charge)
- C3 hull: an arrangement stays if, at some Fermi level in the gap, its lowest formation energy is within hull_margin of its group's
- C3 duplicates: one that relaxed into a lower one's structure goes
- C3 properties (optional), met by a charge state stable somewhere in the gap: require_gap_states, require_spin, require_bound
- D6: the keep_structures lowest structures of each charge state

D Ground and excited states

D_states/ · the survivors, in the electronic cell

(C)

Search with a model of each defect? (model)
- no
- yes

D1 Training data
each defect's C5 geometry steps and rattled pristine cells, in the electronic cell; filtered, a fifth kept for testing

D2 Hybrid single points
HSE06 (norm-conserving) on about a tenth of them, in the electronic cell: first, last and an even sample

D3 Train a model per defect
MACE from scratch: pbe and hse heads on one representation, the total charge embedded (every charge state)

D4: accurate? (test errors; the hybrid's forces at its minima)
- no → Its minima's single points join the data; train again (at most model_rounds)
- yes

D5 Search, electronic cell
ShakeNBreak distortions (±0.7) and rattles in every charge state, relaxed with the hse head; each minimum tried in the other charge states

D6 Rank the ground states
the model's lowest minima, or (no model) C5's, sharing distortions; keep_structures go on

D8 Validate
the screening cell vs bigger cells; Γ vs k-points (for you)

D7 Excited states (ΔSCF)
electronic cell, Γ: ground states relaxed; triplets only; localised levels (participation numbers); ZPL, E_{abs}, ΔQ, 1D S and DW

Band states: pending, for a hybrid later (F3 lists them)

A charge state passes? (ZPL ≥ excite_min, still on the defect)
- no → Dropped unless native (why: summary.txt)
- yes

(E)

D7: which excitations go on
- levels: bands localised on the defect (participation number at most local_fraction of the atoms), in the gap or up to excite_reach inside a band, where PBE's narrow gap pushes many
- the excitation: in each spin channel, the highest occupied level to the lowest empty one; CASTEP's ΔSCF from the ground state's .check, relaxed
- 1D estimates: $E_{rel} = E_{abs} - ZPL$, $\hbar\omega = \hbar v(2E_{rel})/\Delta Q$, $S = E_{rel}/\hbar\omega$, $DW = e^{-S}$ (a lower bound)
- a defect that passes goes on with all its charge states; native defects always do

Test D7 on one defect first

CASTEP's ΔSCF must work on your build: in CaO, Sb_{Ca}–V_O at q = −1 has a ZPL of about 0.53 eV (ADAQ). The thresholds are loose on purpose: the energies are PBE's.

E Production ground and excited states

E_production/ · the electronic cell, every charge state

(D)

E1 Production relaxations
every charge state of each defect D passes on, at A1's settings, in the electronic cell; the pristine cell too

Did a job fail?
- yes → Flagged, left out of F: fix it and rerun with --resubmit-failed
- no

E2 The ground states again
production settings, the bands' weights: the same excitation, found again

Its bands keep their numbers at every k-point? (spread < band_spread_tol)
- no → Use a Γ-only ground state instead, at the production cut-off
- yes

E2 Production ΔSCF
relaxed: ZPL, E_{abs}, ΔQ, 1D S and DW, with D7's verdicts; the excited structures go on to G

(F)

How the CASTEP jobs run

Every CASTEP step hands its jobs to supercell_convergence.py:
- one short dry run per distinct input sizes the jobs: cores or nodes, memory, partition and walltime; where nodes are shared and charged by the core (Cirrus), each job asks for just the cores it needs
- no more than max_jobs are queued at once
- a job that runs out of time carries on from its .check (up to max_restarts times; screening jobs don't restart)
- with tolerate_failures (the screening, hybrid and production steps), a failed job is flagged and the others carry on
- a finished job whose inputs have since changed stops the run, so results from two different settings never mix
- touch <workdir>/STOP stops dwf.py; queued jobs keep running

Stopping and rerunning

<workdir>/state.json records each finished step, with a fingerprint of its settings, input files and earlier results.
- stopped (STOP, Ctrl-C, --nos-wait, a crash)? Run the same command again: finished steps are skipped, and jobs already submitted or finished are picked up
- changed a setting? The steps that depend on it run again; jobs whose inputs are unchanged are reused
- --redo STEP: start that step again (its old files are kept)
- --status: what has finished; --until B: stop after phase B
- --prepare-only: write the job inputs, but submit nothing

Charge corrections: where they're applied

Lany–Zunger (default) or Makov–Payne: a point charge q screened by ε_0 (A2.3), in each cell's own shape (charge_correction).
- C3: the hull, stable charge states and binding (screening cell)
- D8: each size's $E_D - E_{host} + E_{corr}$
- F1–F2: the electronic cell

Not included: potential alignment, an anisotropic ε (averaged).

Why E2 checks the band numbers

CASTEP's ΔSCF constraints name a band number and spin channel, the same at every k-point. A level that disperses, or crosses a host band somewhere, would move the electron into different states at different k-points. In the electronic cell the band edges fold onto Γ and a localised level is flat, so the check passes; where it fails, E2 runs at Γ, where the bands are unambiguous.

Why PBE from E on

CASTEP's ΔSCF does not support hybrid functionals, so E2 and G stay at PBE (A1's settings). The hybrid is used where it decides the structure (D1–D5); the finalists from G go on to better methods (MBPT, or a hybrid ΔSCF elsewhere).

F Thermodynamics

F_thermo/ · every defect that reached E

(E)

F1 Charge corrections
Lany–Zunger or Makov–Payne (charge_correction), in the electronic cell

F2 Formation energies
at each chemical-potential limit, and the transition levels; plotted in formation.png

Find the Fermi level? (fermi_temperature)
- yes → The self-consistent Fermi level of each dopant set (native, each dopant, each mix); quench_temperature: also after a quench
- no

F3: an E2 excitation whose charge state forms in the gap?
- no → End: formation energies, levels and Fermi levels
- yes

(G)

G Optical lineshapes

G_lineshape/ · the excitations F3 sends on

(F)

G1 Configuration coordinate
ground-state and ΔSCF single points along the line between the relaxed states; parabolas: ħω and S of each, and the 1D lineshape

Multi-mode lineshape? (lineshape)
- no → End: 1D lineshapes
- yes

G2 A PBE model per defect
trained on displaced single points (production settings) and E1's steps; one head, the charge embedded

G2 Phonons in a big cell
finite differences with the model; ΔF (or ΔR) on each mode: the partial Huang–Rhys factors S_k

G2 Multi-mode lineshape
S(ħω) and the generating function, at lineshape_temperature

End: lineshapes

Phase G, and after it

PBE lineshapes in the equal-mode approximation (Alkauskas et al. 2014; Razinkovas et al. 2021). The finalists then go on to better methods, outside this workflow.